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1 CREATE DATABASE Customers_Orders_Products;
2 USE Customers_Orders_Products;
3
4 -- Create the Customers table
5 CREATE TABLE Customers (
6     CustomerID INT PRIMARY KEY,
7     Name VARCHAR(50),
8     Email VARCHAR(100)
9 );
10
11 -- Create the Orders table
12 CREATE TABLE Orders (
13     OrderID INT PRIMARY KEY,
14     CustomerID INT,
15     ProductName VARCHAR(50),
16     OrderDate DATE,
17     Quantity INT,
18     FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
19 );
20
21 -- Create the Products table
22 CREATE TABLE Products (
23     ProductID INT PRIMARY KEY,
24     ProductName VARCHAR(50),
25     Price DECIMAL(10, 2)
26 );
27
28 -- Insert records into the Customers table
29 INSERT INTO Customers (CustomerID, Name, Email)
30 VALUES
31 (1, 'John Doe', 'johndoe@example.com'),
32 (2, 'Jane Smith', 'janesmith@example.com'),
33 (3, 'Robert Johnson', 'robertjohnson@example.com'),
34 (4, 'Emily Brown', 'emilybrown@example.com'),
35 (5, 'Michael Davis', 'michaeldavis@example.com'),
36 (6, 'Sarah Wilson', 'sarahwilson@example.com'),
37 (7, 'David Thompson', 'davidthompson@example.com'),
38 (8, 'Jessica Lee', 'jessicalee@example.com'),
39 (9, 'William Turner', 'williamturner@example.com'),
40 (10, 'Olivia Martinez', 'oliviamartinez@example.com'),
41 (11, 'James Anderson', 'jamesanderson@example.com'),
42 (12, 'Kelly Clarkson', 'kellyclarkson@example.com');
43
44 -- Insert records into the Products table
45 INSERT INTO Products (ProductID, ProductName, Price)
46 VALUES
47 (1, 'Product A', 10.99),
48 (2, 'Product B', 8.99),
49 (3, 'Product C', 5.99),
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50 (4, 'Product D', 12.99),
51 (5, 'Product E', 7.99),
52 (6, 'Product F', 6.99),
53 (7, 'Product G', 9.99),
54 (8, 'Product H', 11.99),
55 (9, 'Product I', 14.99),
56 (10, 'Product J', 4.99),
57 (11, 'Product K', 3.99),
58 (12, 'Product L', 15.99);
59
60 -- Insert records into the Orders table
61 INSERT INTO Orders (OrderID, CustomerID, ProductName, OrderDate, Quantity)
62 VALUES
63 (1, 1, 'Product A', '2023-07-01', 5),
64 (2, 2, 'Product B', '2023-07-02', 3),
65 (3, 3, 'Product C', '2023-07-03', 2),
66 (4, 4, 'Product A', '2023-07-04', 1),
67 (5, 5, 'Product B', '2023-07-05', 4),
68 (6, 6, 'Product C', '2023-07-06', 2),
69 (7, 7, 'Product A', '2023-07-07', 3),
70 (8, 8, 'Product B', '2023-07-08', 2),
71 (9, 9, 'Product C', '2023-07-09', 5),
72 (10, 10, 'Product A', '2023-07-10', 1),
73 (11, 11, 'Product D', '2023-07-10', 3),
74 (12, 12, 'Product E', '2023-07-11', 6),
75 (13, 5, 'Product G', '2023-07-12', 2),
76 (14, 4, 'Product H', '2023-07-13', 4),
77 (15, 6, 'Product I', '2023-07-14', 3);
78
79 select * from Customers;
80 select * from Orders;
81 select * from Products;
82
83 ----- ↗
84 ----- ↗
85 -----
86
87
88 --Task 1 :-
89 --1. Write a query to retrieve all records from the Customers table..
90 select * from Customers;
91
92 --2. Write a query to retrieve the names and email addresses of customers
93 whose names start with 'J'.
94 select Name,Email from Customers where Name like 'J%';
95
96 --3. Write a query to retrieve the order details (OrderID,
97 ProductName,Quantity) for all orders..
98 select OrderID,ProductName,Quantity from Orders;
99
100 --4. Write a query to calculate the total quantity of products ordered.
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95 select Sum(quantity) as TotalQuantity from Orders;
96
97 --5. Write a query to retrieve the names of customers who have placed an order.
98 select Distinct C.Name from Customers C inner join Orders O on
    C.CustomerID = O.CustomerID;
99
100 --6. Write a query to retrieve the products with a price greater than $10.00.
101 select * from Products where Price > 10;
102
103 --7. Write a query to retrieve the customer name and order date for all orders placed on or after '2023-07-05'.
104 select C.Name,O.OrderDate from Customers C inner join Orders O on
    C.CustomerID = O.CustomerID where O.OrderDate >= '2023-07-05'
105
106 --8. Write a query to calculate the average price of all products.
107 select avg(Price) as AvgPrice from Products;
108
109 --9. Write a query to retrieve the customer names along with the total quantity of products they have ordered.
110 SELECT C.Name,SUM(O.Quantity) AS TotalQuantity
111 FROM Customers C INNER JOIN Orders O ON C.CustomerID = O.CustomerID
112 GROUP BY C.Name;
113
114 --10. Write a query to retrieve the products that have not been ordered.
115 SELECT P.ProductName FROM
116 Products P LEFT JOIN Orders O ON P.ProductName = O.ProductName WHERE
    O.ProductName IS NULL;
117
118 -----
119 -----
120 -----
121
122 --Task 2 :-
123 --1. Write a query to retrieve the top 5 customers who have placed the highest total quantity of orders.
124 SELECT TOP 5
125     C.CustomerID,
126     C.Name,
127     SUM(O.Quantity) AS TotalQuantity
128 FROM Customers C
129 INNER JOIN Orders O
130 ON C.CustomerID = O.CustomerID
131 GROUP BY C.CustomerID, C.Name
132 ORDER BY TotalQuantity DESC;
133
134 --2. Write a query to calculate the average price of products for each product category.
135 SELECT ProductName,AVG(Price) AS AveragePrice FROM Products
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133 GROUP BY ProductName;
134
135 --3. Write a query to retrieve the customers who have not placed any orders.
136 SELECT C.CustomerID, C.Name FROM
137 Customers C LEFT JOIN Orders O ON C.CustomerID = O.CustomerID
138 WHERE O.CustomerID IS NULL;
139
140 --4. Write a query to retrieve the order details (OrderID, ProductName,
    Quantity) for orders placed by customers whose names start with 'M'.
141 SELECT
142     O.OrderID,
143     O.ProductName,
144     O.Quantity
145 FROM Customers C INNER JOIN Orders O ON C.CustomerID = O.CustomerID
146 WHERE C.Name LIKE 'M%';
147
148 --5. Write a query to calculate the total revenue generated from all orders.
149 SELECT SUM(O.Quantity * P.Price) AS TotalRevenue
150 FROM Orders O INNER JOIN Products P ON O.ProductName = P.ProductName;
151
152 --6. Write a query to retrieve the customer names along with the total revenue generated from their orders.
153 SELECT C.Name, SUM(O.Quantity * P.Price) AS Revenue
154 FROM Customers C INNER JOIN Orders O ON C.CustomerID = O.CustomerID
155 INNER JOIN Products P ON O.ProductName = P.ProductName
156 GROUP BY C.Name;
157
158 --7. Write a query to retrieve the customers who have placed at least one order for each product category.
159 SELECT CustomerID
160 FROM Orders
161 GROUP BY CustomerID
162 HAVING COUNT(DISTINCT ProductName) =
163 (
164     SELECT COUNT(DISTINCT ProductName)
165     FROM Orders
166 );
167
168 --8. Write a query to retrieve the customers who have placed orders on consecutive days.
169 SELECT DISTINCT A.CustomerID
170 FROM Orders A INNER JOIN Orders B ON A.CustomerID = B.CustomerID
171 AND DATEDIFF(DAY, A.OrderDate, B.OrderDate)=1;
172
173 --9. Write a query to retrieve the top 3 products with the highest average quantity ordered.
174 SELECT TOP 3 ProductName, AVG(Quantity) AS AvgQuantity FROM Orders
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175 GROUP BY ProductName
176 ORDER BY AvgQuantity DESC;
177
178 --10. Write a query to calculate the percentage of orders that have a
    quantity greater than the average quantity.
179 SELECT (COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Orders)) AS
    PercentageAboveAverage FROM Orders
180 WHERE Quantity > (SELECT AVG(Quantity) FROM Orders);
181
```